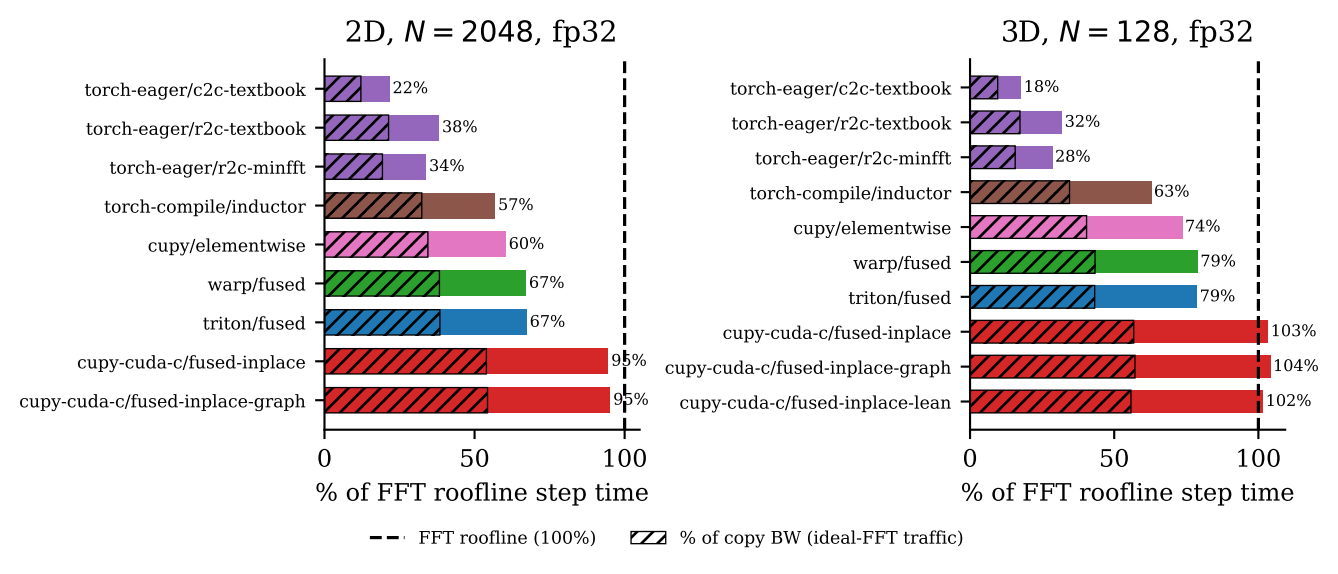
2D, $N = 2048$, fp32
torch-eager/c2c-textbook
torch-eager/r2c-textbook
torch-eager/r2c-minfft
torch-compile/inductor
cupy/elementwise
warp/fused
triton/fused
cupy-cuda-c/fused-inplace
cupy-cuda-c/fused-inplace-graph
22%
38%
34%
57%
60%
67%
67%
95%
95%
0
50
100
% of FFT roofline step time
3D, $N = 128$, fp32
torch-eager/c2c-textbook
torch-eager/r2c-textbook
torch-eager/r2c-minfft
torch-compile/inductor
cupy/elementwise
warp/fused
triton/fused
cupy-cuda-c/fused-inplace
cupy-cuda-c/fused-inplace-graph
cupy-cuda-c/fused-inplace-lean
18%
32%
28%
63%
74%
79%
79%
103%
104%
102%
0
50
100
% of FFT roofline step time
FFT roofline (100%)
% of copy BW (ideal-FFT traffic)